EcoCon-NOS: Leakage-Safe Nested Detector Selection

Training-only detector selection for unseen IoT attack families

STAGE A

A. OUTER LOAFO SETUP

CICIoT2023 modeling pool
Hold out one attack family per outer fold

OUTER TRAIN
Known families only
Train XGBoost + nested detector selection

OUTER CALIBRATION
Known traffic only
Reserved for final calibration

OUTER TEST-KNOWN
Final known-traffic evaluation

OUTER TEST-UNKNOWN
True held-out family
SEALED until final evaluation

STAGE B

B. OUTER TRAIN ONLY - NESTED SELECTION

1. Eligible fine-grained attack labels
from OUTER TRAIN only

2. Create pseudo-unknown folds
up to 12 folds; hold out one eligible fine label

3. Group-safe inner split
80% reference / 20% known validation; hash overlap = 0

4. Evaluate seven candidate detectors
MSP • Entropy • MD-Pred • Min-MD • MDS • CC-kNN • Global-kNN

5. Inner evaluation
known threshold ≈5% rejection; pseudo-unknown UADR + risk

SELECTION RULE
Primary: highest mean inner UADR
Tie <= 0.01: lower risk -> higher worst-fold UADR -> lower predefined cost

SELECT DETECTOR -> FREEZE SELECTION

STAGE C

C. CALIBRATION & FINAL EVALUATION

Frozen selected detector

Refit detector references
using FULL OUTER TRAIN

OUTER CALIBRATION
known traffic only; class-conditional calibration at alpha = 0.05

FREEZE REJECTION RULE

ONLY NOW OPEN FINAL TEST DATA
TEST-KNOWN + TEST-UNKNOWN

Frozen XGBoost + selected detector
Calibrated score / p-value

ACCEPT
known prediction

ABSTAIN / REVIEW
potentially unseen

True outer-unknown data never participate in detector selection or threshold calibration.